HM5V-PD

USB Type-C Power Delivery 3.0 sink controller with I²C interface

Features

- USB PD 3.0 sink; negotiates 5 V, 9 V, 15 V and 20 V profiles
- Up to 100 W (20 V at 5 A with an e-marked cable)
- VBUS over-voltage protection threshold: 22.0 V typical
- I²C control interface, 7-bit address 0x28 (0x29 when ADDR pin is high)
- I²C clock up to 400 kHz (Fast-mode)
- Four general-purpose I/O pins and an active-low INT output
- 10-bit VBUS and IBUS monitoring ADC
- Supply: VDD 3.0 V to 5.5 V
- QFN-24 (4×4 mm) package

General description

The HM5V-PD negotiates power contracts with USB-C sources and reports status over I²C. It drives an external load switch and monitors VBUS voltage and current. An internal comparator disconnects the load switch when VBUS exceeds the over-voltage threshold.

Applications

Laptop docks, Power banks, Battery chargers, Portable test equipment.

Ordering information

Order code	Package	Packing	Temperature grade
HM5VPD-QFN24-T	QFN-24 (4×4 mm)	Tape & reel, 3000	–40 °C to 105 °C

Absolute maximum ratings

Stresses beyond these ratings may cause permanent damage. Functional operation is not implied.

Parameter	Min	Max	Unit
VBUS to GND	–0.3	28	V
CC1, CC2 to GND	–0.3	24	V
VDD to GND	–0.3	6.0	V
Junction temperature TJ	—	150	°C
Storage temperature	–65	150	°C

Recommended operating conditions

Parameter	Min	Typ	Max	Unit
VBUS operating range	4.0	—	21.5	V
Supply voltage VDD	3.0	3.3	5.5	V
I²C clock frequency	—	—	400	kHz
Ambient temperature TA	–40	—	105	°C

Electrical characteristics

Typical values at TA = 25 °C unless otherwise noted.

Parameter	Conditions	Min	Typ	Max	Unit
VBUS over-voltage threshold	VBUS rising	21.3	22.0	22.7	V
VBUS under-voltage lockout	VBUS falling	3.5	3.7	3.9	V
Supply current, attached	VDD = 3.3 V	—	180	260	µA
Supply current, detached	VDD = 3.3 V	—	12	25	µA
ADC resolution	VBUS / IBUS	—	10	—	bit
VBUS measurement accuracy	4 V to 21.5 V	–2	—	2	%
INT output low voltage	IOL = 2 mA	—	—	0.4	V
I²C address (ADDR = GND)	7-bit		0x28		
I²C address (ADDR = VDD)	7-bit		0x29		

Pin configuration and functions

Pin	Name	Type	Description
1	VBUS	Power	USB-C VBUS sense input.
2	CC1	I/O	Configuration channel 1.
3	CC2	I/O	Configuration channel 2.
5	VDD	Power	3.0 V to 5.5 V supply.
7	SDA	I/O	I²C data (open-drain).
8	SCL	Input	I²C clock.
9	ADDR	Input	Address select: GND = 0x28, VDD = 0x29.
10	INT	Output	Active-low interrupt (open-drain).
11	GATE	Output	Drives the external VBUS load switch.

Package and thermal information

Package	Thermal resistance θJA	Dimensions
QFN-24 (4×4 mm)	θJA = 38 °C/W	4.0 × 4.0 mm

Application information

Place a 10 µF ceramic capacitor on VBUS and keep CC1/CC2 traces short and away from switching nodes.

Use pull-up resistors of 2.2 kΩ on SDA and SCL when VDD = 3.3 V at 400 kHz. See AN-107 for bus design guidance.

Revision history

Revision	Date	Change
Rev A	2024-05-10	Initial release
Rev B	2024-12-02	Added ADDR pin; 400 kHz I²C
Rev C	2025-06-20	Tightened OVP threshold: 22.0 V typical (21.3 V min, 22.7 V max)